MIL-S-19500/279C(NAVY)
5 March 1971
SUPERSEDING
MIL-S-19500/279B(NAVY)
14 May 1968

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, TYPES

1N3644, 1N3645, 1N3646 AND 1N3647

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 0.1 ampere silicon rectifiers and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Maximum ratings.

Type	V_R	I_o	I_o T_A = 100°C	i(Surge)	T_{stg}	Altitude
	Vdc	mAdc	mAdc	A	°C	Ft.
1N3644	1500	250	100	14	-65 to +175	40,000
1N3645	2000	250	100	14	-65 to +175	40,000
1N3646	2500	250	100	14	-65 to +175	40,000
1N3647	3000	250	100	14	-65 to +175	40,000

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Diodes shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Design, construction and physical dimensions. Diodes shall be of the design, construction, and physical dimensions shown on Figure 1.

3.2.1 Encapsulating material(s). Encapsulating material(s) used shall be at the device manufacturer's option. Information concerning the material(s) shall be included with the qualification device design data.

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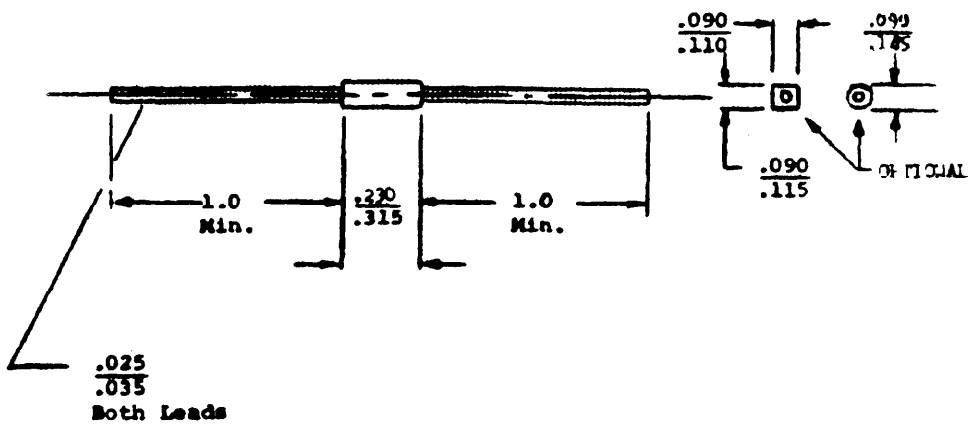


FIGURE 1. Design, construction and physical dimensions of diode, types 1N3644, 1N3645, 1N3646 and 1N3647.

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3.3 Performance characteristics. Performance characteristics shall be as specified in 4.3, 4.4 and 4.5.

3.4 Marking. The following marking specified in MIL-S-19500 may be omitted at the option of the manufacturer:

(a) Manufacturer's identification
(b) Country of origin

3.4.1 Polarity. The polarity shall be indicated by a contrasting color band or dot to indicate the cathode end.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in tables I, II, and III. Group A inspection shall be performed on a sublot basis. Group B inspection, with the exception of subgroups 7 and 8, shall be performed on a lot basis. Group B subgroups 7 and 8 shall be performed on sublots of the highest and lowest voltage types present in the lot. Group C inspection shall be performed on sublots of the highest voltage type for each altitude rating for which qualification applies.

TABLE I. Group A inspection.

Examination or Test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 Method	Specific Conditions			Min	Max	
Subgroup 1			5				
Visual and mechanical examination	2071						
Subgroup 2			5				
Forward voltage	4011	I_F = 250mA		V_F	---	5	Vdc
Reverse current	4016						
1N3644		V_R = 1500Vdc		I_R	---	5	uAdc
1N3645		V_R = 2000Vdc		I_R	---	5	uAdc
1N3646		V_R = 2500Vdc		I_R	---	5	uAdc
1N3647		V_R = 3000Vdc		I_R	---	5	uAdc
Breakdown voltage	4021						
1N3644		BV = 1800Vdc		I_R	---	50	uAdc
1N3645		BV = 2400Vdc		I_R	---	50	uAdc
1N3646		BV = 3000Vdc		I_R	---	50	uAdc
1N3647		BV = 3600Vdc		I_R	---	50	uAdc
Reverse current average	4046	T_A = +100°C					
		I_O = 100mA					
1N3644		V_R = 1050Vac		I_{RO}	---	100	uAdc
1N3645		V_R = 1400Vac		I_{RO}	---	100	uAdc
1N3646		V_R = 1750Vac		I_{RO}	---	100	uAdc
1N3647		V_R = 2100Vac		I_{RO}	---	100	uAdc

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Examination or Test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 Method	Specific Conditions			Min	Max	
Subgroup 1							
Physical dimensions	2066		5				
Subgroup 2			10				
Solderability	2026						
Thermal shock (Temperature cycling)	1051	Cond. C except T High = +175°C					
Thermal shock (glass strain)	1056	Cond. A except T = -40°C T = +120°C					
Moisture resistance	1021						
Subgroup 3			10				
Shock	2016	1,000G 5 Blows X1, Y1, Y2					
Vibration fatigue	2046	Non-Operating 20 G 55 Hz					
Vibration, variable frequency	2056						
Constant acceleration	2006	15,000G X1, X2, Y1					
Subgroup 4			10				
Lead fatigue	2036	Cond. E					
Subgroup 5			10				
Salt atmosphere	1041	48 Hrs.					
Subgroup 6			10				
Surge current	4066	T_A = +25°C I (Surge) = 14A Ten 8 msec surges; 1 surge/min I_O = 250mA					

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Table II. Group B inspection (Continued).

Examination or Test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 Method	Specific Conditions			Min	Max	
Subgroup 7			λ = 10				
High temperature life (non-operating)	1031	T_A = 175°C					
Subgroup 8			λ = 10				
Steady State operation life	1026	T_A = 100°C I_o = 100mAdc					
1N3644		V = 1050Vac					
1N3645		V = 1400Vac					
1N3646		V = 1750Vac					
1N3647		V = 2100Vac					
End points for subgroups 2 through 8							
Forward voltage	4011	I_F = 250mA		V_F	---	5	Vdc
Reverse current	4016						
1N3644		V_R = 1500Vdc		I_R	---	5	uAdc
1N3645		V_R = 2000Vdc		I_R	---	5	uAdc
1N3646		V_R = 2500Vdc		I_R	---	5	uAdc
1N3647		V_R = 3000Vdc		I_R	---	5	uAdc
Breakdown voltage	4021						
1N3644		B_V = 1800Vdc		I_R	---	50	uAdc
1N3645		B_V = 2400Vdc		I_R	---	50	uAdc
1N3646		B_V = 3000Vdc		I_R	---	50	uAdc
1N3647		B_V = 3600Vdc		I_R	---	50	uAdc
Reverse current average	4046	T_A = 100°C I_o = 100mA					
1N3644		V_R = 1050Vac		I_{RO}	---	100	uAdc
1N3645		V_R = 1400Vac		I_{RO}	---	100	uAdc
1N3646		V_R = 1750Vac		I_{RO}	---	100	uAdc
1N3647		V_R = 2100Vac		I_{RO}	---	100	uAdc

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TABLE III. Group C inspection.

Examination or Test	Conditions		LTPD	Symbol	Limits		Unit
	MIL-STD-750 Method	Specific Conditions			Min	Max	
Subgroup 1			10				
Barometric pressure, reduced (altitude operation) (See 4.5.1)	1001	40,000 ft.		---	---	---	---
Measurement during test							
Reverse current	4016						
1N3644		V_R = 1500V					
1N3645		V_R = 2000V					
1N3646		V_R = 2500V					
1N3647		V_R = 3000V					

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Electronic Systems Command. Qualification tests shall consist of the tests specified in 4.2.1. (Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (See 6.1).)

4.2.1 Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III. Group A inspection shall be performed on a sublot basis. Group B subgroups 7 and 8 shall be performed on sublots of the highest and lowest voltage types submitted to qualify those voltage types and all intermediate voltage types. Group C inspection shall be performed on sublots of the highest voltage type for each altitude rating for which qualification is pursued.

4.2.2 Qualification inspection information. When specified in the contract or order, one copy of the qualification inspection data pertinent to the inspection lot shall accompany the shipments. (See 6.3).

4.3 Group A inspection. Group A inspection shall consist of the examinations and tests shown in table I.

4.4 Group B inspection. Group B inspection shall consist of the examinations and tests shown in table II.

4.5 Group C inspection. Group C inspection shall consist of the examinations and tests shown in table III.

4.5.1 Barometric pressure, reduced. The devices shall be suitably mounted in the vacuum chamber and the specified reverse voltage applied (see 1.2). Reverse current shall be monitored. Pressure shall be reduced to the value specified herein and maintained for 30 seconds. Evidence of arcing, corona, or reverse current in excess of 100 microamperes shall be considered a defect.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 The activity responsible for the qualified products list is the NAVAL ELECTRONIC SYSTEMS COMMAND, ELEX 0517, Department of the Navy, Washington, D.C. 20360. Information pertaining to

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qualification of products may be obtained from either the NAVAL ELECTRONIC SYSTEMS COMMAND or the Defense Electronics Supply Center (DESC), Dayton, Ohio 45401, agent for administration of the Qualified Products List. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6". (Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

6.2 Substitution data. Types 1N3643 and 1N3992 have been deleted from this specification. Type 1N3957 of MIL-S-19500/228 may be substituted for type 1N3643. There is no direct replacement for type 1N3992.

6.3 Ordering data.

(a) Qualification inspection information, if required (See 4.2.2).

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
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